

Install web server and setup reverse proxy with NGINX to Python Flask Backend and NodeJS Backend

☒ Step 1: Install Required Packages

```
sudo apt update
sudo apt install nginx python3-pip nodejs npm -y
```

☒ Step 2: Setup Flask Backend (Port 5000)

◇ You can skip the virtual environment part below if any errors occur.

```
mkdir ~/flask-backend && cd ~/flask-backend
python3 -m venv venv
source venv/bin/activate
```

◇ **Mandatory step**

Install Flask:

```
pip install Flask
```

Create `app.py`:

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Flask!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Run the Flask app:

```
python3 app.py
```

☑ Step 3: Setup Node.js Backend (Port 3000)

```
mkdir ~/node-backend && cd ~/node-backend
npm init -y
npm install express
```

Create `index.js`:

```
const express = require('express');
const app = express();

app.get('/', (req, res) => {
  res.send('Hello from Node.js!');
});

app.listen(3000, () => {
  console.log('Node app listening on port 3000');
});
```

Run the Node.js app:

```
node index.js
```

☑ Step 4: Configure NGINX as Reverse Proxy

Create a new NGINX config file:

```
sudo nano /etc/nginx/sites-available/myapp
```

Paste the following:

```
server {
  listen 80;
  server_name localhost;

  location /flask {
    proxy_pass http://localhost:5000;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
  }
}
```

```
location /node {  
    proxy_pass http://localhost:3000;  
    proxy_set_header Host $host;  
    proxy_set_header X-Real-IP $remote_addr;  
}  
}
```

Enable the configuration:

```
sudo ln -s /etc/nginx/sites-available/myapp /etc/nginx/sites-enabled/
```

☒ Step 5: Fix Conflict with Default Config

! Problem:

By default, NGINX uses `/etc/nginx/sites-enabled/default`, which has:

```
listen 80 default_server;
```

This makes it the default handler for all traffic on port 80, including `http://localhost`, so your `myapp` config is ignored.

☒ Fix: Disable the Default Config

```
sudo rm /etc/nginx/sites-enabled/default  
sudo systemctl restart nginx
```

This ensures your `myapp` config becomes the only listener on port 80.

☒ Step 6: Test the Reverse Proxy

- Flask app: <http://localhost/flask>
 - Node.js app: <http://localhost/node>
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